

Spatial Degradation of Hepatocyte Zonation

Using a healthy spatial liver atlas to assign a pseudo-zonal coordinate to spatially-blind diseased liver cells — and then measuring its collapse across the disease course

Project proposal, primer & full work plan

Combining a healthy spatial liver atlas (Yakubovsky et al., Nature 2026)
with a single-nucleus disease cohort (Gribben et al., Nature 2024)

Team of 2 · Two concentrated days + preparation

snRNA-seq

Spatial transcriptomics

Zonation reconstruction

Zone classifier + entropy

FDR

MASLD → MASH → cirrhosis

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Throughout, **COURSE** marks an idea from Computational Genomics 76553 (Ex 1–2, Lec 5, Lec 15–16); **NEW** marks one we add for this project.

1 The question in one breath

A healthy liver is spatially organised: each liver cell sits at a position along a blood-flow gradient and does the metabolic job that suits its position. One recent paper measured that organisation in healthy human liver *with the positions kept*. Another profiled tens of thousands of liver cells across the full fatty-liver disease course, but with a method that *throws the positions away*. This project uses the healthy map as a ruler to assign a *pseudo-zonation coordinate* to each spatially-blind diseased cell, and then measures — rigorously — how that spatial organisation falls apart as the disease worsens, and whether the cells with the most degraded zonal identity are also the ones drifting toward a duct-like (plasticity) program.

Two emphases run through everything below. First, "falls apart" is read from the *population*, never from any single cell's recovered coordinate (which we never treat as a true position): healthy zonation makes the two opposite gene programs mutually exclusive across cells, and de-zonation is when that exclusivity — not a label we assigned — breaks down. Second, the unit of inference is always the **donor** (~47 of them), never the individual cell (~69k) — a discipline we justify in detail when we reach the statistics.

WHY IT CAN SCORE FULL MARKS

It is not a reproduction of either paper — neither paper performs this cross-dataset decoding step. It runs on the exact machinery of this course (single-cell processing and clustering, differential expression, and Benjamini–Hochberg FDR). And it fits two days because every individual operation is standard; the originality is in how they are combined.

1.1 What is already built, so the two days are execution, not setup

All the slow, fiddly **plumbing and de-risking is already done** in preparation. Nothing below is the scientific result — those are produced live during the hackathon — but the scaffolding, the data conversions, and the method's positive control are finished and checked in. This is why two days is realistic.

Area	What is done (prework)	Evidence / file
Data acquired	Both papers' raw data downloaded; Paper 2 supplementary tables + the Paper 2 GitHub repo (landmark genes, zonation scripts) pulled.	data/raw/
Data converted to Python-native	Paper 1 Seurat .rds → counts.mtx (30,117 genes × 69,426 hepatocytes) + metadata; Paper 2 .mat → paper2_train.npz (zone-labelled training nuclei). No R/MATLAB needed again.	data/processed/ paper1/ paper2_train.npz
Signatures built	Both signature schemes are frozen for the head-to-head: the transcriptome-wide program (~1,640 significantly-zonated hepatocyte genes, $q < 0.05$; the default) <i>and</i> the exact Paper 2 landmark set; plus expanded (~100) and curated core sets and a sensitivity set for robustness.	signatures/ (+ README)
Pipeline written	The full donor-level analysis (load → score → validate → collapse → pseudobulk DE → plasticity) is coded and smoke-tested end-to-	src/pipeline.py

end on synthetic donor data — it runs and produces the right shapes.

Method validated (positive control)	On <i>real</i> Paper 2 healthy hepatocytes the coordinate recovers zonation : the pericentral marker rises with it and the periportal marker falls (see "reading the validation plot", §5.5). The ruler works before we ever touch disease data.	results/figures/ p2_validation.png
Repo structured	Clean layout (data/{raw,processed}, src/{prep,steps,plotting}, signatures/, results/); per-step code stubs and plotting machinery (one function per artefact) pre-written; run_all.py driver; GitHub sync set up for uni ↔ home + the two of us.	src/steps/, src/ plotting/, PROJECT_MAP.md
Write-ups	This primer, the slide deck, the professor green-light email, and the work-division plan.	docs/

SO WHAT IS ACTUALLY LEFT FOR THE TWO DAYS?

Only the **science**: harmonize genes (Step 3) → score + train the classifier on real data (Step 4) → validate + ruler battery (Steps 5/5b) → the collapse curve (Step 6, H1) → zone-resolved DE (Step 7, H2) → the plasticity link (Step 8, H3) → optional bonus, figures, write-up. Every step has a stub file waiting and an acceptance check. We deliberately did *not* pre-run the science — that is the hackathon, and doing it live is the point.

2 The biology, from the ground up

2.1 What the liver does, and the two cell types that do it

The liver is the body's central metabolic factory: it buffers blood sugar (storing and releasing glucose), synthesises most plasma proteins such as albumin, makes the bile acids that digest fat, converts nitrogen waste into urea, and detoxifies drugs. The single fact that drives this entire project is that these jobs are **not** shared evenly among liver cells — they are divided up by *where* a cell sits in the tissue.

Two kinds of epithelial cell do that work, and the distinction between them is central to this project. **Hepatocytes** are the bulk "factory" cells — ~80% of the liver's mass — arranged in plates along the blood path; they do essentially all the metabolism, detoxification, plasma-protein synthesis and bile *production*. Their identity marker is albumin (ALB). **Cholangiocytes** are a small minority (a few percent) that line the **bile ducts** — the branching drainage plumbing (the "biliary tree") sitting at the portal corners — and they *transport* the bile hepatocytes make out toward the gut, without doing the bulk metabolism. Their markers are cytokeratins KRT7, KRT19 and SOX9.

Two things to hold onto. **(1) These are different cell types, with normally fixed, non-overlapping identities** — a hepatocyte expresses ALB and not KRT19; a cholangiocyte the reverse. Mature epithelial identity is supposed to be locked, so a cell expressing *both* ("biphenotypic") is a striking sign of identity coming unfastened — that is **plasticity**, and Paper 1's headline finding. **(2) Keep this axis separate from zonation.** Zonation is variation *within* the hepatocyte population (where on the porto–central axis a hepatocyte sits); the hepatocyte ↔ cholangiocyte distinction is a difference *between cell types*. They are two different axes — which is exactly why "de-zonation" (losing your *position*, still a hepatocyte) and "plasticity" (losing your *cell-type* identity) are distinct things our analysis measures with *separate* signatures (§2.5, §5.4).

What flows in and out. The liver has a **dual blood supply**, both entering at the portal triads: the **portal vein** (~75% of inflow) carries *nutrient-rich, only moderately oxygenated* blood drained from the gut, spleen and pancreas — so everything absorbed from digestion passes through the liver first; the **hepatic artery** (~25%) brings *oxygen-rich* blood from the aorta. The two mix in the sinusoids, flow past the hepatocytes to the **central vein**, and leave via the hepatic veins to the heart. So periportal ≈ "just in from the gut," pericentral ≈ "on the way back out to the body." **Bile** — the greenish fluid hepatocytes secrete (bile acids/salts, bilirubin, cholesterol) that emulsifies dietary fat and carries away waste — drains the *opposite* way: from tiny **canaliculi** between hepatocytes into the bile ducts of the portal triads, then onward to the gallbladder and duodenum. Cholangiocytes are the cells that line those ducts.

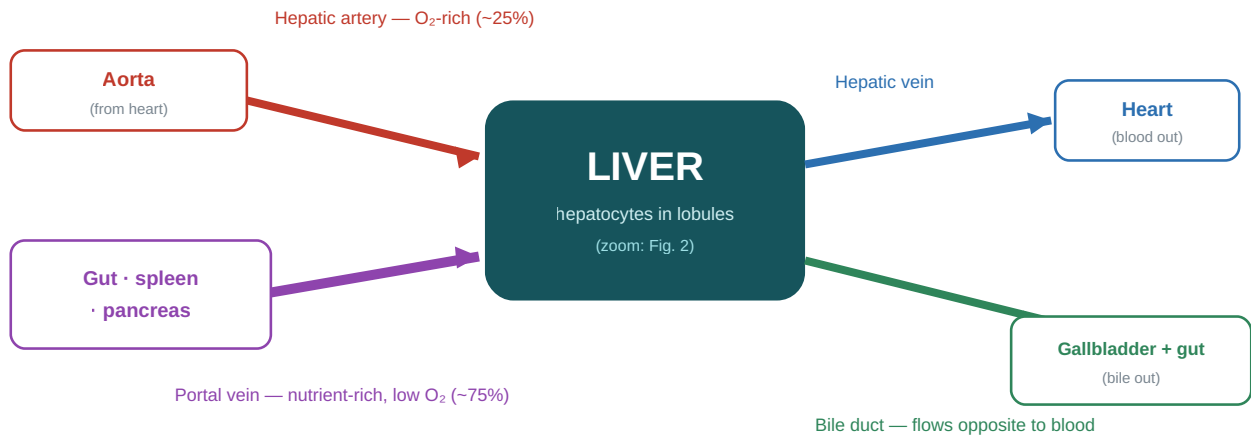


Fig. 1 — The liver in its plumbing. Two supplies enter together at the portal triads — the **portal vein** (from gut, spleen, pancreas; nutrient-rich, low O_2) and the **hepatic artery** (O_2 -rich, from the aorta) — mix through the sinusoids and leave as blood via the **hepatic vein** to the heart. **Bile** exits the other way, through the bile ducts to the gallbladder and gut. Fig. 2 zooms into one lobule.

2.2 The lobule, the gradient, and zonation

The liver's repeating unit is the **lobule**, a roughly hexagonal tile about 0.5–1 mm across. Blood enters at the corners through the **portal triads** and flows inward to a **central vein** at the centre. Along that journey — the **porto–central axis** — the blood gives up its oxygen and nutrients, so a cell's chemical environment depends entirely on how far along the axis it sits: relatively oxygen-rich at the portal edge (where the oxygenated hepatic artery feeds in), oxygen-poor at the centre.

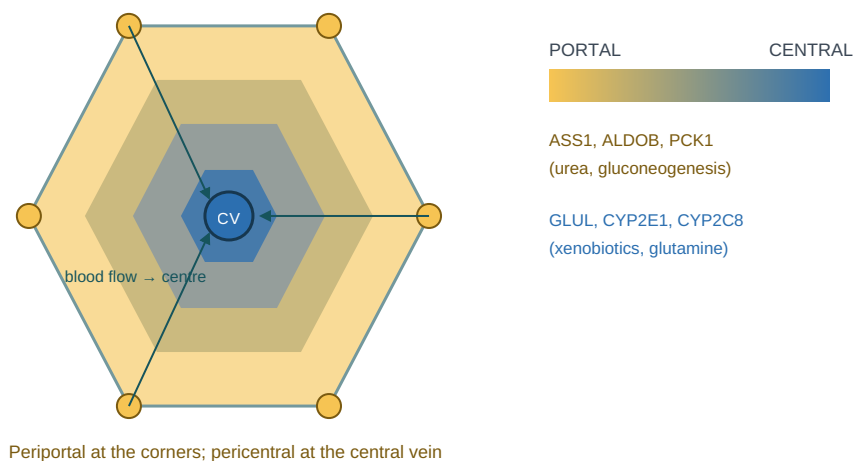


Fig. 2 — The lobule. Blood flows inward from the oxygen-rich portal corners (yellow) to the oxygen-poor central vein (blue); zonation runs along that path. The bands are *schematic*: real zonation is a smooth gradient along the sinusoids (the acinus zones 1–3), not discrete rings, and bile drains the opposite way (toward the portal triads). Representative marker genes for each end are shown.

Because the environment changes smoothly along the axis, hepatocytes specialise smoothly too. This division of labour is called **liver zonation**. Broadly, the **periportal** end runs much of the urea cycle (ASS1) and gluconeogenesis (PCK1, ALDOB), while the **pericentral** end runs glutamine synthesis (GLUL) and cytochrome-P450 **xenobiotic metabolism** (CYP2E1, CYP2C8). A key *human-specific* result of Paper 2 is that several functions long

assumed periportal are actually **pericentrally shifted** in humans — the gluconeogenic gene PCK2 and urea-cycle enzymes ASL/OTC/NAGS among them, while ASS1 stays periportal — so we take the marker sets *empirically from the data*, not from the mouse textbook. Crucially the cell carries out this specialisation by *switching genes on and off* in proportion to its position. So a hepatocyte's set of active genes is, in effect, a readout of where it sits. **That is the hinge of the whole project**: if position is encoded in expression, then position can be decoded back out of expression — even after the physical location is gone.

2.3 The disease that degrades it: the MASLD spectrum

Both papers study **MASLD** (metabolic dysfunction-associated steatotic liver disease, formerly NAFLD), the commonest chronic liver disease, which advances through ordered stages:

Stage	What happens	Hallmark
Healthy	Normal architecture and zonation	baseline
MASLD / steatosis	Fat droplets accumulate inside hepatocytes	steatosis
MASH	Fat + inflammation + hepatocyte injury (ballooning)	inflammation
Cirrhosis	Scarring (fibrosis) replaces working tissue	fibrosis
End-stage	Architecture destroyed; regenerative nodules	organ failure

Terminology: MASLD/MASH is the current name; the Paper 1 dataset and older literature label the same stages **NAFLD/NASH**. We use MASLD/MASH in prose but keep the dataset's own labels when reading its metadata.

Why it happens (in brief). MASLD is a disease of metabolic overload. Chronic excess calories and **insulin resistance** push more fatty acids into hepatocytes than they can burn or export, so triglyceride accumulates as droplets — **steatosis**. When the lipid load outstrips the cell's capacity it turns toxic (*lipotoxicity*): mitochondrial and ER stress, reactive-oxygen damage, hepatocyte injury and death. That damage recruits immune cells (inflammation = **MASH**) and activates hepatic stellate cells to lay down collagen scar (**fibrosis** → **cirrhosis**). Genetics (e.g. PNPLA3, TM6SF2) and the gut–metabolic milieu tune individual risk.

IS DE-ZONATION A MECHANISM OR A SYMPTOM?

Mostly a **readout**: zonation is maintained by spatial signalling gradients (notably Wnt/ β -catenin and oxygen), and metabolic stress plus disrupted Wnt degrade it — so de-zonation largely *reports* how far cellular identity has unravelled rather than causing the disease. But it is not cosmetic: the two ends run *different* jobs (pericentral drug-detox vs periportal urea/gluconeogenesis), so losing the gradient means **specific liver functions fail in a specific order**. Why we map it: (1) a quantitative **severity/progression marker** finer than histological staging; (2) it pinpoints **which functions degrade first** (e.g. pericentral CYP drug metabolism — directly relevant to drug dosing in liver disease); (3) it may flag **early, pre-fibrotic** damage; (4) it connects to the **plasticity** route linked to cancer risk.

2.4 The two ways identity breaks — and why keeping them apart is the whole point

As disease advances, the orderly lobule degrades in two *distinct* ways. **De-zonation** is loss of normal *zonal restriction* within the hepatocyte population: genes that should be confined to one end of the porto–central axis

become **blurred, silenced, or co-expressed with the opposite program**. (Co-expression of opposite ends is one important mode; global loss of a program and flattening of the gradient are others — §5.6 tells them apart.) **Epithelial plasticity** is loss of identity *across* cell types: the boundary between hepatocyte and cholangiocyte identity blurs, producing **biphenotypic** cells (bi = two, phenotype = identity) that express markers of *both* lineages. This is molecular, not physical — no cell crawls into a duct; the gene-expression programs mix. (Paper 1 reports such transdifferentiation with organoid + PI3K–AKT–mTOR evidence; the in-vivo directionality is complex, so we treat it as boundary-blurring, not a clean one-way hepatocyte → cholangiocyte switch.)

So the contrast is clean: de-zonation = a hepatocyte forgetting *where on the axis* it belongs; plasticity = a hepatocyte forgetting *that it is a hepatocyte at all*. They look like two depths of the same unravelling — which is why we measure the first quantitatively across the disease course and test whether it travels with the second.

DE-ZONATION VS PLASTICITY — THEY SOUND IDENTICAL BUT AREN'T (AND THIS IS H3)

Both are "a cell expressing two things it shouldn't at once," so they blur together easily. The difference is **which two things**, i.e. *which boundary* is being crossed:

	De-zonation	Plasticity
Boundary crossed	<i>Position</i> within the hepatocyte (portal vs central end of the axis)	<i>Cell type</i> identity (hepatocyte vs cholangiocyte)
Co-expressed pair	pericentral + periportal programs (e.g. CYP2E1 + ASS1) in one cell	hepatocyte + bile-duct programs (e.g. ALB + KRT7) in one cell
Still a hepatocyte?	Yes — it just lost its <i>address</i> on the assembly line	Drifting — it is forgetting its <i>profession</i> altogether

Analogy: a factory worker who forgets *which station* they man (de-zonation) is still a factory worker; one who forgets they *work in this factory at all* (plasticity) is something else. **H3 asks whether these are the same workers** — do the hepatocytes that have most lost their *position* tend to be the same ones drifting toward *bile-duct identity*? De-zonation is the milder, earlier slip; plasticity the deeper one — H3 tests whether losing your address predicts losing your job.

THE NEED THIS SETS UP (READ THIS BEFORE THE NEXT SECTION)

To measure de-zonation we would need two things we do not obviously possess together: (1) a trustworthy, quantitative map of **healthy** zonation — a ruler — and (2) a large population of **diseased** cells, spanning every stage, to lay against that ruler. The next section shows that exactly these two things exist — in two separate papers — and that nobody has put them together.

3 The data that exists — and the lucky gap

Our need has two halves — a healthy ruler and diseased cells — and there happen to be two papers, each supplying exactly one half. But to see why they fit together (and why neither alone is enough) we first need one idea about *how* the measurements are made.

3.1 Two ways to measure a tissue: keep the map, or dissolve it NEW

In ordinary sequencing, a molecule's physical position on the instrument is meaningless — genomic identity comes only from alignment. The whole idea of spatial transcriptomics is to make a position mean something biological again: not on the flow cell, but in the tissue. There are two families of technique, and the difference between them is our opportunity.

- **Single-nucleus RNA-seq (snRNA-seq)** *dissolves the tissue first*. Its flow: lyse the cells and release nuclei into a **suspension**; the nuclei, now floating in a tube, are pooled and shuffled; each is captured in its own droplet with a barcoded bead; reverse-transcribe, amplify, sequence, and end up with a nuclei×genes count matrix. The reward is enormous — you read essentially the whole transcriptome of each cell, and nuclei survive the freezing and fibrosis that would destroy whole cells, which is the only reason late-stage scarred tissue can be profiled at all. The price is irreversible: the moment nuclei go into suspension, **which lobule or axis-position each came from is gone forever**. (scRNA-seq is the whole-cell cousin; both are dissociated methods.)
- **Spatial transcriptomics** measures expression *without dissociating* — on an intact tissue section, so each measurement carries its (x,y) coordinate. The workhorse, **10x Visium**, lays the slice on a slide printed with a grid of barcoded **spots** (~55 µm, a few cells each); every transcript inherits its spot's barcode, so position is retained. Higher-resolution variants — **Visium HD** (a fine 2–8 µm grid), **MERFISH** (imaging that counts a few hundred chosen genes molecule-by-molecule), and **PhenoCycler** (cyclic antibody imaging of proteins) — trade transcriptome breadth for resolution. Because position is kept, you can compute how any gene varies along the porto–central axis — i.e. build the zonation ruler.

No method gives everything: spatial methods sit on a triangle of competing goods — transcriptome breadth, spatial resolution, and sensitivity — and you choose your corner to match the question. For our purposes the headline trade is simple: dissociated = all genes, no position; spatial = position kept, with some cost in breadth or sensitivity.

TWO DIFFERENT "LOSSES" — KEEP THEM SEPARATE, THEY MATTER ALL THE WAY THROUGH

(1) **Technical** loss of coordinates: snRNA-seq dissolves the tissue, so we no longer know where a cell sat. This is a limitation of the *instrument*. (2) **Biological** de-zonation: in disease the cell itself loses its zonal identity. This is the *phenomenon we study*. The whole project is: use a spatial reference to undo loss (1) just well enough to measure loss (2).

THE MOVE AT THE HEART OF THE PROJECT: SPATIAL AS A RULER

A spatial dataset is a reference map in which expression is tied to position — for the liver, where each expression profile sits along the porto–central axis. We **learn** from that reference the relationship "this expression pattern → this zone-index." Then we take the dissociated disease data, which on its own has no coordinates, and **apply** that learned mapping back onto it to decode an inferred position per cell. The spatial map is the ruler; the dissociated data are the measurements we re-place onto it. The spatial reference effectively donates its coordinate system to data

that has none — which is what lets us ask spatial questions ("is zonation disrupted in disease?") on a dataset with no spatial information of its own.

3.2 Paper 2 — the healthy ruler

Yakubovsky, Bahar Halpern, Shir et al. (Itzkovitz lab). *A spatial atlas of the healthy human liver from live donors*. Nature (2026). doi:10.1038/s41586-026-10377-y.

This paper supplies our ruler. Its key move was to obtain genuinely healthy human liver from **living** donors (ordinary "normal" references come from brain-dead organ donors, whose livers are metabolically disturbed), and to profile 16 samples with the full spatial battery plus snRNA-seq. From the spatial data it reconstructed a quantitative porto–central **zone index** per spot and identified the genes that vary most cleanly along the axis.

WHAT TO READ & EXTRACT FROM PAPER 2

Figure 2 is the one that matters — it is the zonation reconstruction. Panel 2a shows Visium slides coloured by the spot zone index from portal to central; panel 2b is the ranked set of zonated genes. The paper analyses **1,724 hepatocyte-specific genes**, of which **1,141 show significant zonal differences** ($q < 0.25$), each with a gene-zonation score (Fig. 2b), and normalizes expression as z-scores of $\log_{10}$ UMI-sum-normalized values — match that normalization so the signatures transfer. Read the Methods paragraph on **zonation reconstruction** / "axis spot annotation" (they seed the axis using landmark genes) to understand how the zone index is defined. Concretely, pull from this paper: (i) the **ranked zonated-gene table** → your transcriptome-wide pericentral/periportal signatures; (ii) the **landmark genes** anchoring each end (pericentral CYP2E1, CYP2C8, GLUL, and the human-specific PCK2; periportal ASS1, ALDOB, PCK1) → the faithful landmark baseline; (iii) the location of its **snRNA-seq** data and the script that assigns zonation to those nuclei (your classifier training set, §6).

3.3 Paper 1 — the diseased cells

Gribben, Galanakis, Calderwood et al. (Vallier, Mohorianu, Allison labs). *Acquisition of epithelial plasticity in human chronic liver disease*. Nature (2024). doi:10.1038/s41586-024-07465-2.

This paper supplies the cells to test. It profiled **47 liver biopsies** by snRNA-seq across the entire spectrum — healthy (n=4), MASLD/NAFLD (n=7), MASH (n=27), cirrhosis (n=4), end-stage (n=5) — yielding ~100,000 QC-passing nuclei including **69,426 hepatocytes**. Its headline biology is transdifferentiation between hepatocytes and cholangiocytes, driven by PI3K–AKT–mTOR; for us, the relevant secondary observation is that hepatocytes lose zonation — but it reports this only qualitatively.

WHAT TO READ & EXTRACT FROM PAPER 1

Figure 1c–e defines the experiment and the cell-type annotation: the snRNA-seq workflow schematic (nuclei isolated by *tissue lysis and FACS sorting* from snap-frozen biopsies; 10x; CellRanger v5.0.0), the overall annotated UMAP (1d), and the marker bubble-plot (1e) you use to **confirm which cells are hepatocytes** (ALB). **Figure 2** is their zonation evidence — 2a is the hepatocyte UMAP split by stage, and 2b is a correlation of pericentral vs periportal marker expression across disease progression — precisely the qualitative result we turn quantitative and continuous. Extract: (i) the **cell-type label** field and the **stage** field from the object's metadata; (ii) their pericentral (GLUL, AXIN2) and periportal (ASS1, HAL) markers; (iii) the **plasticity / biphenotypic markers** for our H3 (KRT7, KRT19, SOX9, plus SOX4, KRT23, NCAM1). One data caveat to honour: the GEO record notes samples GSM6112262 and GSM6112263 had swapped metadata (fixed June 2025) — use the corrected version.

3.4 The gap — they are mirror images neither paper has joined

	Paper 2 (the ruler)	Paper 1 (the cells)
Spatial coordinates	Kept (Visium, MERFISH)	Destroyed (snRNA-seq)
Disease coverage	Healthy + early steatosis only	Full spectrum, healthy → end-stage
Zonation	Quantitative, per-spot ruler	Noted only qualitatively
Scale	44,243 spots (+ snRNA-seq)	69,426 hepatocytes

One paper has the **map** but no disease; the other has the **disease** but no map. Both are recent — Paper 1 (2024), Paper 2 (2026) — and each was built to answer its own question, so no one has yet used the healthy spatial ruler to decode the disease cohort. That untaken step is our project.

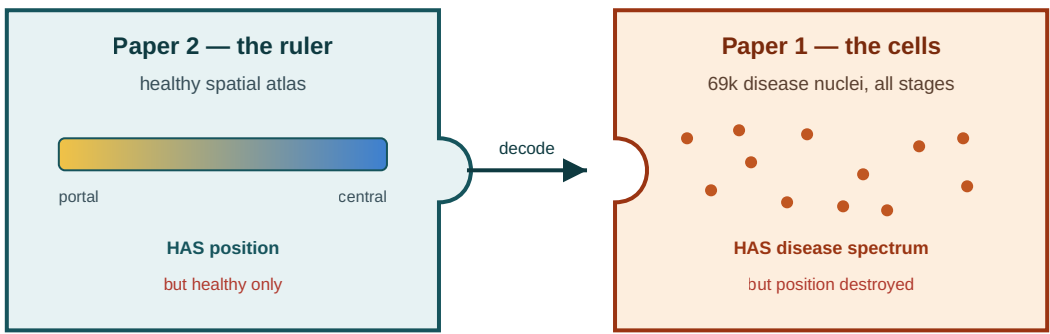


Fig. 3 — Two halves of one puzzle. The healthy atlas (left) carries a quantitative portal → central position but no disease; the single-nucleus cohort (right) spans the whole disease course but has thrown its positions away. The project's one move is to fit the left piece's ruler into the right piece's cells — using the healthy gradient to assign each diseased nucleus a pseudo-zonation coordinate.

"BUT PAPER 1 ALREADY SHOWED ZONATION LOSS" — QUALITATIVE VS QUANTITATIVE VS CONTINUOUS

Paper 1's zonation evidence (their Fig. 2) is a UMAP, a correlation of two marker genes, and **microscopy images** of a handful of marker proteins (GLUL, ASS1) in healthy vs end-stage tissue. That is **qualitative** (you look and see it's messier), **discrete** (two extremes), and **narrow** (a few markers). We add three things it lacks: (1) **quantitative** — a de-zonation *number* per stage with confidence intervals and a significance test, so we can say *how much* and *whether it's real*; (2) **continuous** — every one of the 69,426 hepatocytes gets a real-valued position on the portal ↔ central axis (not a portal/central bucket), tracked across *all five* stages as a curve, not two snapshots; (3) **transcriptome-wide** — the position is read from the whole zonation program (the ~1,141 significantly zonated genes among 1,724 hepatocyte-specific genes) via an external healthy reference, not 2–3 markers by eye. In one line: they showed *that* zonation is lost; we measure *how much*, *where on the axis*, and *with what significance*, across the entire disease course.

4 The precise question

RESEARCH QUESTION

Using the healthy porto–central zonation ruler from Paper 2, can we assign a quantitative pseudo-zonation coordinate to every spatially-blind hepatocyte in Paper 1, and so measure **how, and how much, hepatocyte zonation degrades as MASLD advances from healthy to end-stage** — and is that degradation linked to the transdifferentiation Paper 1 reported?

WHY THIS MATTERS — BEYOND "WE MADE A MAP"

De-zonation is not a curiosity; it is **liver function failing in a readable order**, because the two ends of the axis do *different jobs*. Concretely:

- **Drug safety.** The pericentral zone runs cytochrome-P450 **drug metabolism** (CYP2E1/CYP3A4). If pericentral identity blurs *first* (H2), drug clearance changes early in disease — directly relevant to **dosing and drug toxicity** in liver patients.
- **Ammonia / brain.** The periportal zone runs the **urea cycle** (ASS1) that clears ammonia. Losing it drives **hyperammonemia and hepatic encephalopathy**. So *which* zone erodes *when* predicts *which* failure mode comes first.
- **A finer severity readout.** Histological staging is coarse and subjective. A continuous, molecular **de-zonation score** could grade severity more finely and may flag **early, pre-fibrotic** damage before histology does.
- **Cancer link.** The de-zonation → plasticity axis (H3) connects to **de-differentiation**, a recognised route toward hepatocellular carcinoma.
- **A reusable instrument.** Once validated, the reference-decoder reads zonal identity out of the *many* existing single-nucleus disease atlases that have **no** spatial data — value far beyond this one dataset.

In one line: we build a quantitative readout of *how the liver's division of labour disintegrates*, in an order that maps onto real clinical failure modes.

Each hypothesis below is what we hope to show (the *alternative*); every statistical test silently compares it against its **null**, the boring "no-effect" world, and a p-value measures evidence against that null. (The full statistical toolkit for H1 and H3 is in §5.7; the tests are named here.)

Hypothesis (alternative)	Null (H0) we test against	Test
H1 — monotonic erosion. A de-zonation score rises Healthy → MASLD → MASH → cirrhosis → end-stage.	The score is flat across stages — no association with disease.	Ordered trend (Jonckheere–Terpstra / Spearman on stage rank)
H2 — zone-specific vulnerability. Particular programs (e.g. pericentral CYP detox) collapse earlier/faster.	A gene's zonal pattern does not change across stages.	Pseudobulk DE + BH-FDR
H3 — plasticity link. The most de-zonated hepatocytes are the ones gaining cholangiocyte identity.	De-zonation and plasticity scores are independent.	Within-donor correlation / Mann–Whitney

4.1 H1 — does zonation erode as disease advances?

Claim: a single de-zonation score climbs steadily Healthy → NAFLD → NASH → cirrhosis → end-stage. The score is a *population* metric — e.g. the weakening of pericentral–periportal mutual exclusivity, or the narrowing of the coordinate's spread — **not** any single cell's position (see §5.4). **Null (H0):** the score is flat — unrelated to stage.

The test, plainly. Our stages have a natural order, so we ask whether the score moves *monotonically* along it. **Spearman correlation (ρ)** measures whether two things rise or fall together by rank: $\rho=+1$ perfectly increasing, -1 perfectly decreasing, 0 no relationship — here, score vs stage-rank. The **Jonckheere–Terpstra** test is the dedicated significance test for an *ordered* trend across ordered groups; it returns one p-value for "the score increases step by step across the stages." (Spearman gives the strength; Jonckheere gives the formal yes/no.) **What counts as a pass:** a significant trend ($\rho>0$, $p<0.05$) that survives **donor-level** resampling, so it isn't driven by a single patient. **If it fails (still useful):** a flat score means zonation is surprisingly robust; a rise-then-plateau means the damage saturates early.

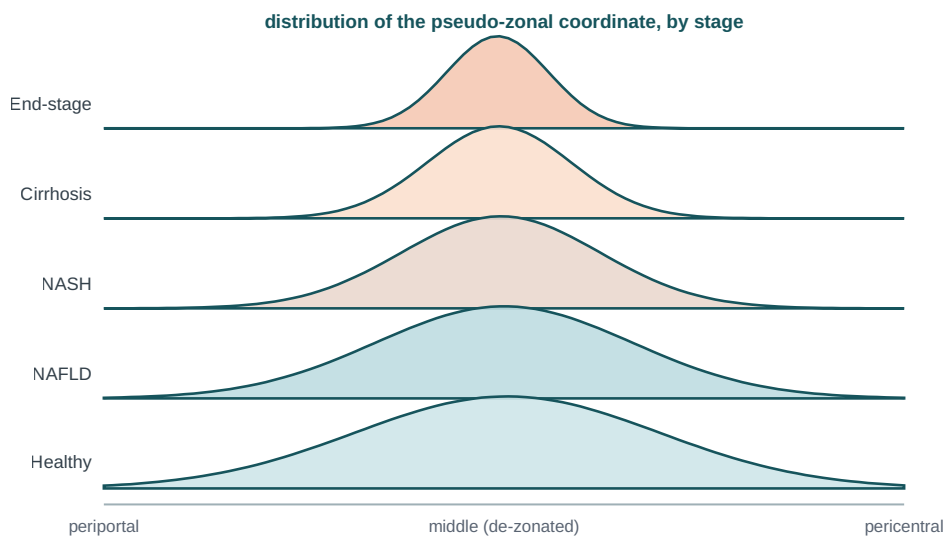


Fig. 4 — What H1 should look like (illustrative). Zonation is a **continuum**, so the healthy coordinate is a **single wide hump** (cells occupy every position along the axis) — *not* two peaks. The signal of healthy zonation is the **width** (and that markers track the axis, Fig. 5), not bumpiness. As disease advances the hump **narrows** toward the middle: cells stop committing to either end.

H1 measures that narrowing (coordinate spread / variance) plus the weakening pericentral–periportal anti-correlation — one number per donor, tested across stages. A genuinely de-zonated liver would show a *narrow* central spike; a healthy one a *broad* spread.

4.2 H2 — which gene programs collapse, and where on the axis?

Claim: the collapse is not uniform — specific programs (e.g. pericentral CYP detox genes) lose their zonal pattern earlier/faster. **Null (H0):** a given gene's zonal pattern does not change across stages. **The test, plainly.** We bin cells into pseudo-zones (portal / mid / central) and, within a zone, test each gene for change across stages — **differential expression (DE)**, run at the donor level by pseudobulk (§5.7). Testing ~20,000 genes at $p<0.05$ would give ~1,000 false hits from noise alone, so we apply **Benjamini–Hochberg FDR**, which turns p-values into q-values that cap the expected fraction of false positives among the genes we call significant. **What counts as a pass:** a coherent set of genes at $q<0.05$ whose loss explains the collapse, ideally enriched for known zoned programs. **If it fails (still useful):** no significant genes means the collapse is diffuse rather than program-specific — itself a statement about *how* zonation breaks.

4.3 H3 — do the de-zonated cells become the plastic ones?

Claim: the hepatocytes that have lost zonal identity are the same ones drifting toward cholangiocyte identity (gaining KRT7/KRT19). **Null (H0):** de-zonation and plasticity are independent. **The test, plainly.** We score each hepatocyte for a plasticity signature, split cells into "de-zonated" vs "still-zonated," and ask whether the de-zonated group has higher plasticity. The **Mann–Whitney U** test (a.k.a. Wilcoxon rank-sum) compares two groups by pooling and ranking all values and seeing whether one group sits at the top — no bell-curve assumption. Crucially we test this **within stage / within donor** (per-donor correlations, or a model $\text{plasticity} \sim \text{de-zonation} + \text{stage} + \text{donor}$) — *not* pooled across all cells, where late disease simply having more of both would fake a link (the stage confound spelled out in §5.7). **What counts as a pass:** de-zonated hepatocytes show significantly higher plasticity ($p < 0.05$) and the two scores correlate positively within donors. **If it fails (still useful):** independence would mean losing zonal identity and gaining duct identity are separate processes — also worth knowing.

Why this is not reproduction. Neither paper computed a cross-dataset zonation coordinate for the disease cohort, a stage-resolved collapse curve, or the de-zonation $\leftrightarrow$ plasticity link. We use Paper 2 strictly as a *decoder* for Paper 1. And a **negative result is still a result:** if the collapse is non-monotonic, or driven by only some programs, or if the ruler itself dissolves at a stage, each is a real, defensible finding — because the method and its controls are the deliverable. The lasting value is threefold and none of it is made redundant by a future real-spatial study: a reusable **reference-based decoder** for the many dissociated atlases that have no spatial data; a **mechanistic ordering** of which zonal programs lose legibility first (H2); and a **principled failure analysis** of when and how the healthy program stops being legible (Step 5b), tied to plasticity (H3).

5 From data to a position-number: the computational logic

§2 promised that position is written into expression. This section turns that promise into a procedure, building it one forced step at a time: each tool appears only because the previous step created a problem that demands it. Nothing here is a tool for its own sake.

5.1 Where we begin: the data table COURSE

After sequencing, the data is one giant table: a row per gene, a column per cell, each entry the number of molecules (UMIs) of that gene caught in that cell. Tens of thousands of each; everything downstream is arithmetic on this table. Reaching it is the alignment of Exercise 1 (placing reads on the genome) and the molecule-counting of Exercise 2 (deduplicating PCR copies into unique counts) — but the published datasets hand us the finished table, so that is our starting line.

5.2 The table lies — so first we clean it (QC + normalization) COURSE

Two flaws make the raw table unsafe. **First, not every column is a real cell.** Empty droplets, debris, and dying or leaky nuclei produce junk columns; if we keep them they later masquerade as "confused" cells and fake the very signal we hunt. So we discard them — filtering on too-few molecules, too-few detected genes, or too-high mitochondrial content (a dying-cell tell). This is **quality control (QC)**. **Second, the surviving numbers aren't comparable:** a cell sequenced twice as deeply shows twice the counts for a purely technical reason. So we rescale every cell to a common depth, then take a logarithm — so that a *doubling* counts the same whether a gene goes $100 \rightarrow 200$ or $1,000 \rightarrow 2,000$, which is the multiplicative scale biology actually lives on (we use $\log(1+x)$, because $\log(0)$ is undefined and the table is full of zeros). This is **normalization**.

QC WE INHERIT; NORMALIZATION WE REDO — EXACTLY WHY

Paper 1's authors already cell-filtered their published object, so we **inherit their QC** (plus a one-line check that the kept cells are ALB-expressing hepatocytes); we do *not* re-filter from scratch. But we **re-normalize from the raw counts** (CP10k $\rightarrow$ log1p $\rightarrow$ z-score), for three reasons. **(1) A cross-dataset transfer needs identical preprocessing.** We score Paper 1 against signatures derived from Paper 2, and apply a classifier trained on Paper 2 — a score only means the same thing on both sides if both datasets went through the *same* normalization, so we run one pipeline over both, starting from counts. **(2) Their processed matrix is gene-incomplete.** A Seurat object carries multiple slots: the raw counts slot holds *every* gene (~30k), while the `scale.data` slot usually holds only the ~2,000 highly-variable genes (covariates regressed out) — so several zonation signature genes may be absent or differently scaled there. We pull counts, not their processed values. **(3) Reproducibility.** A transparent pipeline we control can be described and defended in the report, instead of inheriting an undocumented, software-version-specific one.

5.3 The zeros are a trap — and the trap dictates the method COURSE

Most of the table is zeros, and a zero is ambiguous: the gene may be genuinely off, or on-but-missed. Only a fraction of a cell's molecules are ever captured, so a gene present in a few copies frequently reads zero by pure chance — the same Poisson sampling you met modelling library complexity in Exercise 2 (the probability of seeing

none is $e^{-\lambda}$, large when the expected count λ is small). The missed-but-present zero is called **dropout**, and snRNA-seq is especially prone to it because a nucleus holds little RNA. The conclusion is decisive and shapes everything after it: **no single gene's value in a single cell can be trusted.**

5.4 Therefore: read position from many genes at once (signature scoring)

NEW

Because one gene is unreliable but the zonation *program* spans many genes (Paper 2 finds ~1,141 significantly zoned hepatocyte genes, §3.2), we read position from *many* genes pooled together — which also averages out the random per-cell dropout. A **signature score** is the (normalized, scaled, background-subtracted) *mean* expression of a chosen gene set. "**Background-subtracted**" means: from each cell's mean over the signature genes we subtract that same cell's mean over a matched set of random *control* genes, so the score reflects the *signature specifically* rather than how highly-expressing / deeply-sequenced the cell is overall (exactly what Scanpy's `score_genes` does). We build a **pericentral** and a **periportal** signature from Paper 2's ruler and define a single coordinate per cell:

$$\text{zonation_coord} = \text{score}(\text{PERICENTRAL}) - \text{score}(\text{PERIportal})$$

High = pericentral-like, low = periportal-like; a middle value is *ambiguous* — it can mean "expresses neither" or "expresses both," so a single cell's middling coordinate is only a *candidate* for de-zonation, confirmed by the population-level structural metrics below, not by the coordinate alone. The coordinate is a *signed real number* (sign = which end; magnitude = how strongly committed), centred near zero — **not** a 0–1 probability; probabilities come later, from the classifier (§5.5). (Aside on the supporting cast: **PCA, UMAP and clustering** [COURSE](#) are used only as scaffolding here — to isolate the hepatocytes and to draw a score-coloured map — because Paper 1 already provides cell-type labels. The result comes from the coordinate, not the picture.)

THE SIGNATURE, RUN BOTH WAYS — LANDMARK AND TRANSCRIPTOME-WIDE (A DELIBERATE STRENGTHENING)

We compute the coordinate with **two** gene schemes and report both, checking they agree:

- **(a) The exact Paper 2 landmark set** — the verbatim 20+20 pericentral/periportal landmark genes Paper 2 used to seed its axis. This is *faithful to the original plan*, whose zonation inference was marker/landmark-based, and it is the interpretable sanity baseline.
- **(b) The full transcriptome-wide program** — the ~1,640 significantly-zoned hepatocyte genes ($q < 0.05$, well-expressed, present in Paper 1). This is the **default**, and is a *deliberate strengthening* of the landmark plan: it reads position from the *entire* program rather than a handful of markers by eye, which is the whole motivation of the project and is also what makes the non-circular held-out-gene test (Step 7) meaningful.

Two further sets bridge them as robustness checks — a ~100-gene expanded set (landmark $\cup$ core $\cup$ top-ranked) and a curated 13/8-gene **core**. **The honest framing:** transcriptome-wide is the headline because it is stronger, but we run the landmark set alongside and show the collapse is the *same* either way — so the result is not an artefact of choosing a large gene set. (An earlier draft mistakenly made the 20-gene landmark set the default, which silently contradicted the transcriptome-wide motivation; corrected here.)

In code: `src/steps/step4_score.py` — `coord = mean_z(pericentral) - mean_z(periportal)`; default gene sets are `signatures/*_full.txt` (the **transcriptome-wide program**, via `config.DEFAULT_SET="full"`); switch with `config.signature_files("paper2_landmark"/"expanded"/"core")` for the landmark baseline and robustness.

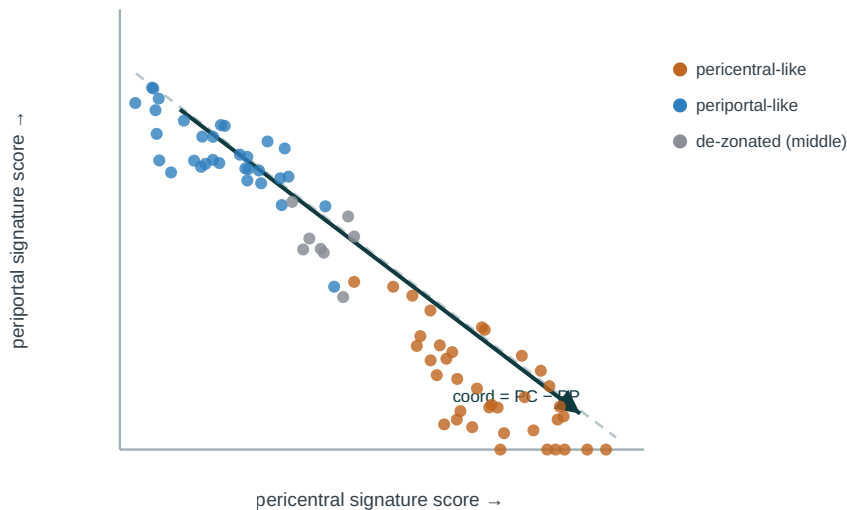


Fig. 5 — How one coordinate is read from two scores. Each dot is a cell, placed by its pericentral score (x) and periportal score (y). In healthy tissue the two are **mutually exclusive** — cells fall into two anticorrelated clouds (dashed guide), and a single number, $\text{coord} = \text{PC} - \text{PP}$ (the dark arrow), captures where each sits on the portal ↔ central axis. The grey "de-zonated" cells in the middle — high on neither or muddled on both — are exactly what proliferates as disease erases the separation (Fig. 4).

HONESTY GUARDRAIL

The ruler is a **1D** axis, so we recover a **pseudo-zonation coordinate**, never a physical (x,y) location. We say "reconstructed zonation coordinate," not "predicted spatial position."

WHAT "DE-ZONATION" MEANS PRECISELY — WHY IT ISN'T CIRCULAR, AND WHAT IT CAN'T SEE

We define de-zonation **operationally**, as the degradation of three properties of the cell *population* — not of any single cell's coordinate, which we never treat as a true position:

- **(1) Mutual exclusivity.** In healthy liver a hepatocyte is high-pericentral *or* high-periportal, so the two scores are strongly *anti*-correlated across cells. De-zonation = cells expressing *both* at once → that anti-correlation weakens.
- **(2) Spread.** Healthy hepatocytes span the whole axis, so the coordinate is a *wide* distribution (typically a single broad hump — a continuum, not two peaks); collapse bunches cells toward the middle (narrow). We track variance/IQR — the *width* is the signal, regardless of the hump's exact shape.
- **(3) Internal coherence.** Do the pericentral genes still co-vary with *each other* — is the program still a coherent module? (Step 5b.)

Why this isn't circular: assigning each cell a coordinate from its expression does *not* predetermine any of (1)–(3) — a disease that preserved zonation would keep strong anti-correlation and wide spread, both fully allowed outcomes. The 1-D coordinate alone is ambiguous (a middle value = "expresses neither" *or* "expresses both"), which is exactly why the honest metrics are these structural ones.

What this can not see (honest scope): we measure loss of the *expression* program, not physical re-arrangement. If diseased cells kept their gene-identity but were shuffled in space (a periportal-type cell sitting where pericentral cells belong), our method — like any dissociated snRNA method — would report *no* change, because the positions were never recorded. Architectural scrambling with intact expression is out of scope; detecting it needs spatial data on the diseased tissue (the very thing that does not yet exist).

5.5 A stronger reader: a zone classifier, and why its confusion is the signal

NEW

The signature score (§5.4) is robust but blunt — it gives one number and no sense of how *sure* we are. A classifier adds exactly that. **What it is, here:** we show a model many hepatocytes whose zone is already *known* ("training"), and it learns the mapping from a cell's expression to its zone. Then, for a brand-new cell, it outputs not a single label but a **probability for each zone** — e.g. [central 0.92, mid 0.06, portal 0.02].

Why the probabilities are the whole point. The *shape* of that probability vector measures confidence, and lost confidence is de-zonation. A healthy, clearly-zonated cell gives a **peaked** vector (the model is sure: 0.92 central). A de-zonated cell — now part-pericentral, part-periportal — gives a **flat** vector (0.4 / 0.35 / 0.25): the model genuinely can't tell, because the cell no longer commits to a zone. We compress "peaked vs flat" into one number, the **Shannon entropy** of the vector, $H = -\sum p_k \log p_k$ — maximal when probability is spread evenly, zero when one zone holds all the mass (low = confident, high = confused). The **mean entropy per disease stage is our headline de-zonation meter**, and watching each zone's confidence separately tests whether pericentral identity blurs before periportal (H2).

In code: `src/steps/step4b_classifier.py` — calibrated multinomial logistic regression trained on `paper2_train.npz`, held-out confusion matrix on Paper 2 first, then `predict_proba` on Paper 1 → per-cell entropy = $-\sum p \cdot \log p$.

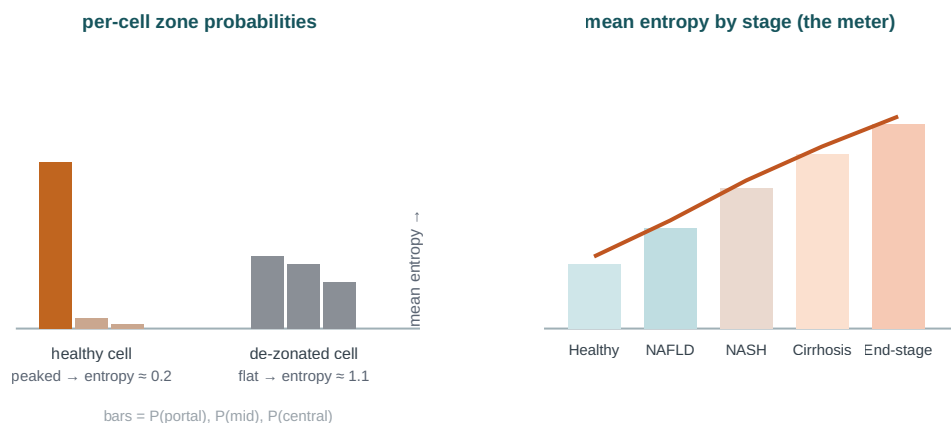


Fig. 6 — Why the classifier's *confusion* is the signal (illustrative). Left: a healthy hepatocyte yields a **peaked** probability vector (the model is sure of its zone → low entropy); a de-zonated cell yields a **flat** one (the model genuinely can't tell → high entropy). Right: averaging that entropy per stage gives the headline de-zonation meter — expected to climb as identity blurs. Per-zone entropy (not shown) tests whether pericentral confidence falls first (H2).

THE CATCH — AND THE TRICK THAT FIXES IT (SAME PLATFORM, BORROWED LABELS)

The catch. A classifier can "cheat" by latching onto *technical* quirks of whatever data it trained on instead of real biology. Paper 2's main data are **Visium spots** — small multi-cell tissue patches, a different technology. If we trained on those spots and asked the model about Paper 1's **single nuclei**, much of what it "learned" would be "Visium-vs-single-nucleus" technical difference.

The trick. Paper 2 *also* ran **snRNA-seq** — the *same* platform as Paper 1. So we train on Paper 2's snRNA-seq nuclei and apply to Paper 1's snRNA-seq nuclei: the same measurement type on both sides, so the model can't win by reading the technology — what carries over is genuine zonation.

But where do the labels come from, if Paper 2's snRNA also lost its coordinates? This is the crux. The labels do

not come from the snRNA nuclei's own (lost) positions. Paper 2 profiled the *same livers two ways* — spatially (Visium, positions kept) *and* by snRNA-seq. It used the **spatial** data to define the porto–central zones, then **transferred those zone labels onto its snRNA nuclei** (mapping each nucleus to a zone via shared expression; their repo's `parse_snRNAseq_combined_atlas.m`). So the training set is snRNA *expression* (same platform as Paper 1) paired with a *zone label borrowed from the spatial modality* — labelled data after all. **Our current fallback** (until we port that mapping) labels each Paper 2 nucleus by the signature-tercile rule from §5.4; that makes the classifier a sanity baseline, and swapping in the spatial-derived labels is the upgrade that makes it ground-truth.

A free sanity check. Scoring and the classifier are two independent routes to the same answer. On *healthy* cells, where zonation is intact, they should agree — and that agreement is built-in evidence the signal is real biology, not an artefact of either method.

ENTROPY IS AN INTERPRETATION, NOT A DIRECT MEASUREMENT — HIGH ENTROPY ≠ AUTOMATICALLY DE-ZONATION

A confused (high-entropy) prediction means "the model can't tell which zone" — but de-zonation is only *one* reason a cell might confuse the model. The others:

- **Too little data in the cell** — a poorly-sequenced nucleus (few molecules / few detected genes) simply doesn't give the model enough to decide: a *technical* shrug, not lost zonation.
- **A cell unlike anything it was trained on** ("out-of-distribution") — e.g. a stressed or dying cell. The model was only taught healthy zones, so anything weird comes out uncertain regardless of zonation.
- **Patient/batch differences** — systematic technical differences between donors or runs shift expression and muddy the calls.
- **A poorly-calibrated model** — some models report wishy-washy probabilities even when they should be sure; then high entropy is the model's fault, not the cell's.
- **Loss of hepatocyte identity altogether** (not just *zonal* identity) — a deeper phenomenon that also makes zone prediction hard.

So we only call high entropy "de-zonation" when it is **not** explained by those — i.e. it survives checking against the cell's QC metrics, its donor/batch, and the module-level diagnostics of §5.6. Practically, we use a **calibrated** classifier (multinomial logistic regression) so the entropy means what we think it does.

5.6 The honesty problem: when does the ruler stop being a ruler? NEW

Here is the project's deepest risk. The ruler is calibrated on healthy tissue, but disease may corrupt the very genes it relies on. A coordinate collapsing to the middle could mean (a) genuine de-zonation, (b) the marker genes globally crashing for unrelated reasons, so the tick-marks fall off the ruler, or (c) the signature dissolving into noise. The raw number can't tell these apart — so we measure the *structure* of the signature per stage: do the pericentral genes still co-vary *with each other* (is the ruler intact)? does the pericentral–periportal anti-correlation across cells weaken (true de-zonation, robust to global shifts)? do two random halves of the gene set give the same coordinate (reproducibility)? and is a module switched *off* (program lost) versus still on but co-expressed with its opposite (positional restriction lost)? These diagnostics are Step 5b, and they let us say which kind of breakdown we are seeing rather than assuming the convenient one.

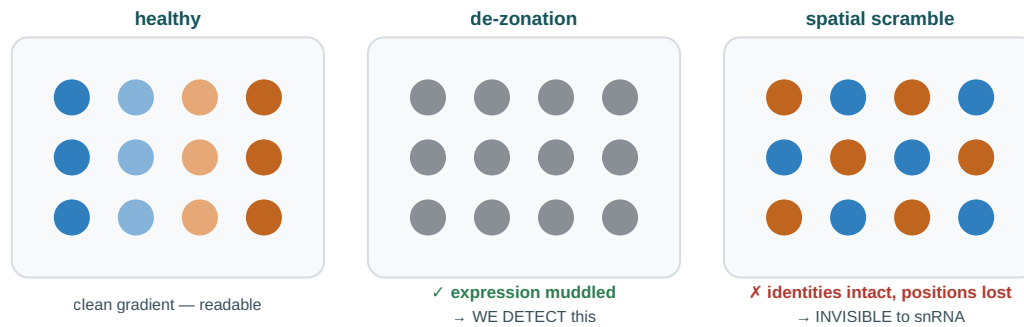


Fig. 7 — What the method can and cannot see. Each dot is a cell, coloured by expressed identity (blue = periportal, orange = pericentral, grey = both/neither). We detect de-zonation when the **expression program** degrades (middle: cells stop committing) — that is real and measurable. We are **blind** to pure spatial scrambling (right): if cells kept distinct identities but were physically rearranged, a dissociated method records no change, because the positions were never measured. This honest scope limit is why a future spatial-disease study would complement, not duplicate, our result.

5.7 Comparing stages without fooling ourselves (the statistics) COURSE

Every test below shares one governing rule, so we state it once and apply it everywhere.

THE ONE RULE THAT GOVERNS EVERYTHING: THE DONOR IS THE UNIT OF INFERENCE

We have ~47 donors. Each contributes thousands of cells, from which we compute *one* summary per donor (e.g. a de-zonation score). The biological question is a statement about **donors**, so the sample size that matters is **n ≈ 47**, *not* ~69k cells. If you run a test on cell-level values, you silently inflate n from 47 to ~10⁵; but cells from one donor are not independent — they share that donor's genetics, treatment, batch and biology. The p-value becomes absurdly tiny and meaningless. This error is **pseudoreplication**. The fix is a two-step pattern used throughout: **(1) collapse** cells to one number per donor; **(2) run the donor-level test**. When we resample for confidence intervals, we resample *donors*, not cells, for the same reason. (DE, in §below, achieves this by *pseudobulk*: aggregating to donor×zone before testing.)

H1 — the ordered-trend tests. The stages are *ordered*, and ordering is information: a test that uses it is more powerful than one that ignores it. **Spearman's ρ** is Pearson's correlation computed on the *ranks* of stage and score, so it detects any monotone relationship regardless of exact spacing or outliers ($\rho=+1$ perfectly increasing, 0 no trend). The **Jonckheere–Terpstra (JT)** test is the purpose-built ordered cousin of Kruskal–Wallis: its null is "all stage distributions identical," its alternative the directional "medians don't decrease as you walk along the stages." It builds its statistic from the Mann–Whitney counts of every ordered stage-pair and spends all its power on the one pattern we predicted in advance — a march in stage order — so when that ordering is real it detects it with a smaller sample (the cost: a non-monotone, up-then-down effect can slip past it). For the confidence interval we use a **donor-level bootstrap** (resample the ~47 donors with replacement, recompute ρ , take the 2.5/97.5 percentiles); if it excludes 0 the trend is significant in the bootstrap sense, and the width reports precision honestly. And the negative control is a **label-shuffle permutation**: keep the scores, randomly reassign stage labels across donors, recompute ρ thousands of times to build the empirical null. A valid pipeline run on shuffled labels lands the observed statistic in the bulk of that null ($p \approx 0.5$) — a pipeline that "finds" a trend in shuffled data is broken.

H3 — association, with the stage confound removed. The trap: *both* de-zonation and plasticity tend to rise with stage, so a naive pooled correlation across all cells looks positive even if the two are unrelated within any single

donor — a textbook Simpson's-paradox confound. The whole job is to remove it, never to pool. We compute the correlation **inside each donor** (where stage is held fixed and cannot drive a spurious link), then aggregate the ~47 within-donor ρ_d values and test whether their centre sits above zero (sign test / Wilcoxon signed-rank), reporting both the mean ρ (strength) and the fraction of donors with $\rho_d > 0$ (consistency). The regression equivalent is OLS $\text{plasticity} \sim \text{dez} + C(\text{stage}) + C(\text{donor})$: the $C(\text{donor})$ term makes each comparison within a donor, and the coefficient β on dez is the partial association after stage and donor are removed (clustered standard errors or a donor random effect keep that p-value honest — otherwise it is pseudoreplication in regression clothing). Where we contrast two groups directly (de-zonated vs zonated), the **Mann–Whitney U** test pools and ranks all values and asks whether one group hogs the high ranks; its effect size is the rank-biserial r (equivalently $2 \cdot \text{AUC} - 1$), which we always report beside the p-value.

H2 — differential expression with FDR. Once every hepatocyte has a coordinate, we bin cells into pseudo-zones and ask, within a zone, which genes change across disease stage — this is **differential expression**, the gene-level story behind the collapse. The danger is volume: testing ~20,000 genes at $p < 0.05$ yields ~1,000 false positives from noise alone (the "throw enough nets and you catch plastic bottles" problem of Lecture 5). The **Benjamini–Hochberg (BH)** procedure controls the *false discovery rate* — the expected fraction of false positives *among the genes we call significant* — rather than forbidding any false positive at all (which Bonferroni's α/m does, at the cost of losing nearly all real signal at this scale). BH sorts the m p-values, finds the largest rank k with $p_{(k)} \leq (k/m) \cdot q$, and declares those significant; the associated **q-value** of a gene is the smallest FDR at which it would be called. Everything we report as "significant" is at **q < 0.05**, never a raw p-value — and always paired with an **effect size** (log fold-change, or the coordinate×stage slope of Step 7), because a tiny q says a change is *real*, not that it is *large*.

In code: `src/steps/step7_de.py` — `pseudobulk (donor×zone)`, per-gene interaction `expr ~ coord + stage + coord:stage`, `multipletests(method='fdr_bh')`; see the algorithm block in §6 Step 7.

READING THE VALIDATION PLOT: WHY A SINGLE CENTRAL PEAK IS STILL ZONATION

When we ran the ruler on Paper 2's healthy hepatocytes, the coordinate came out as **one broad hump centred near zero** — not two peaks — and that worried us at first. It shouldn't: **a single wide hump is exactly what intact zonation looks like.**

- **Zonation is a gradient, not a switch.** Cells sit at *every* position from portal to central, so the coordinate is a *continuum*; a histogram of a continuum is a single spread-out hump. You would only see two spikes if hepatocytes came in two discrete states — which is not the biology. So unimodal $\neq$ "no zonation."
- **The proof is in the slopes, not the histogram.** In the same validation figure the pericentral marker *CYP2E1* *rises* with the coordinate and the periportal marker *ASS1* *falls*. That monotonic ordering is the evidence the axis is real zonation; with no zonation those scatters would be flat clouds.
- **De-zonation is therefore a change in shape, not in modality.** We never expect "two peaks become one." We expect the one hump to get **narrower** across disease (variance shrinks toward the middle) and the *CYP2E1/ASS1* slopes to **flatten**. So the healthy single-peak-with-correct-slopes result is a *pass*, not a problem.

THE WHOLE LOGIC IN ONE LINE

table → clean it (QC, normalize) → trust no single gene (dropout) → pool genes into a coordinate (signature, both ways) → sharpen with a same-platform classifier (entropy = blur) → check the ruler still holds (5b) → compare stages with donor-level, FDR-controlled tests.

6 The work plan, in detail

6.1 The exact inputs

Raw inputs live in `data/raw/`; the two big ones were **already converted** once into Python-native files in `data/processed/`. **The pipeline reads only the processed files** — it never touches the `.rds` or `.mat` again.

Input	Raw — <code>data/raw/</code>	Converted once → <code>data/processed/</code>	Used for
Paper 1 cells	GSE202379_...AllCells.rds (.gz) · 2.6 GB · Seurat/R	paper1/counts.mtx + genes.txt, barcodes.txt, *_metadata.csv	69,426 disease hepatocytes + per-cell stage & donor
Paper 2 atlas	combined_scrnaseq_atlas_...mat · 2.0 GB · MATLAB v7.3	paper2_train.npz (zone-labelled nuclei)	classifier training set (same platform as Paper 1)
Paper 2 ruler	zon_struct_all_full.mat · 204 MB	— (read directly for the positive control)	per-spot zone index + zonation profiles
Paper 2 signatures	Human-liver/.../Hepatocyte-{PC,PP}-LM.csv + supp. tables	signatures/*.txt (full = transcriptome-wide default + paper2_landmark baseline + expanded / core)	the pericentral / periportal gene sets, both ways

THE TWO CONVERSIONS ARE PYTHON, ONE-OFF, AND ALREADY DONE

(1) **Paper 1**: an R script (`src/prep/01_extract_paper1_hepatocytes.R`) exported the Seurat `.rds` to `counts.mtx` + metadata once — then read in Python with `scipy.io.mmread`. (2) **Paper 2**: `src/prep/02_convert_paper2_mat.py` read the MATLAB `.mat` (v7.3 = HDF5) with `h5py` and saved gene names + expression + zone labels as `paper2_train.npz`. After these two prep steps the analysis is **pure Python on data/processed/**. The signatures need only the landmark CSVs / supplementary table — no `.mat` at all.

6.2 Environment

Python, throughout. `scanpy/anndata` (or plain `scipy.sparse`) for the single-cell handling; `numpy/scipy/pandas/scikit-learn` for normalization, the signature, the classifier and statistics; `statsmodels` for `multipletests(method='fdr_bh')`; `h5py` to read Paper 2's `.mat` once; `matplotlib/seaborn` for figures. Where the course expects understanding (normalization, FDR, the classifier), we implement it explicitly rather than hide behind a one-liner.

6.3 The steps

Every step below already has a **stub file** (signature + docstring with algorithm, I/O and acceptance check) and a **plotting function** waiting; the donor-level reference implementation for the statistical steps lives in `src/pipeline.py`. The hackathon is filling these in against real data.

Step	Code stub	Plot	Artefact / answers
1 Ruler	<code>prep/02_convert_paper2_mat.py;</code> <code>signatures/</code>	<code>plot_a1_reference_profiles</code>	A1 signatures + training set
2 Load/QC	<code>steps/step2_load_qc.py</code>	—	A2 matrix + counts
3 Harmonize	<code>steps/step3_harmonize.py</code>	—	A3 gene map
4 Score	<code>steps/step4_score.py</code>	<code>plot_a4_coordinate</code>	A4 coordinate
4b Classifier	<code>steps/step4b_classifier.py</code>	—	A4 zone probs + entropy
5 Validate	<code>steps/step5_validate.py</code>	<code>plot_a5_validation</code>	A5 positive control
5b Ruler diag.	<code>steps/</code> <code>step5b_ruler_diagnostics.py</code>	<code>plot_a5b_ruler</code>	A5b validity
6 Collapse	<code>steps/step6_collapse.py</code>	<code>plot_a6_collapse</code>	A6 · H1
7 Zone DE	<code>steps/step7_de.py</code>	<code>plot_a7_volcano</code>	A7 · H2
8 Plasticity	<code>steps/step8_plasticity.py</code>	<code>plot_a8_plasticity</code>	A8 · H3
9 Bonus	<code>steps/step9_bonus_enrichment.py</code>	—	A9 enrichment

Step 1 — Build the ruler from Paper 2

Inputs: Paper 2 zonated-gene table (supplement) and/or `zon_struct_all_full.mat`; for 1c also `combined_scrnaseq_atlas_M5M6M7M8.mat` + `parse_snRNAseq_combined_atlas.m`.

1a — signatures, both ways. Load the zonated-gene table; build the **transcriptome-wide** set (all genes with $q < 0.05$, well-expressed, present in Paper 1, sided by Center-of-Mass; the default), and freeze the **exact Paper 2 landmark** set verbatim as the baseline; intersect both with Paper 1 genes (record how many survive). **Reserve a random held-out subset** of the transcriptome-wide zonation genes (e.g. 30%) that builds *neither* the coordinate nor the classifier — the independent set for the non-circular collapse test in Step 7.

1b — sanity-plot. Plot the zonation profile of the top ~10 genes each end to confirm monotonic, opposite gradients (catches sign/mapping errors early).

1c — classifier training set. Run / port `parse_snRNAseq_combined_atlas.m` to attach a spatial-derived zone label (or zone index binned into 3–5 classes) to each Paper 2 nucleus; export an (expression, zone_label) table.

Decision: if porting the MATLAB is slow, fall back to signatures only (1a–1b) for Day 1 and add the classifier later.

Artifact A1: signature lists (full + landmark) + zonation-profile figure + zone-labelled training table.

Step 2 — Load & QC the Paper 1 hepatocytes

Input: GSE202379_SeuratObject_AllCells.rds.

2a — subset. Read the object; pull `cell_type` and `stage` from metadata; keep cells labelled Hepatocyte; confirm they express ALB and form a coherent cluster (quick PCA/UMAP).

2b — QC (sanity only, not a fresh pipeline). We *inherit* the authors' published QC; we do **not** re-filter from scratch. Do only light **sanity checks** (ALB-positive hepatocytes; per-stage/donor counts) and optional **sensitivity filtering** (e.g. drop the lowest-UMI tail) to confirm robustness. Record anything removed.

2c — normalize. Library-size normalize → log1p → per-gene z-score (§5.2), matching Paper 2's units so signatures transfer. Pull the raw counts slot, not `scale.data`.

2d — bookkeeping. Build a per-stage, per-donor cell-count table (donor IDs needed for the bootstrap in Step 6).

Artifact A2: clean normalized hepatocyte matrix (~69k) + QC plots + per-stage/per-donor counts.

Step 3 — Harmonize genes & fix the batch axis

Inputs: A1 + A2.

3a — ID mapping. Reconcile gene identifiers (symbol vs Ensembl, version drift) between the two datasets; report the mapping rate.

3b — intersect. Restrict signatures and classifier features to the shared gene set; report surviving signature sizes (if a signature shrinks badly, widen the top-N in Step 1).

3c — batch-robust features. Use z-scored or rank-based features so the score measures portal-vs-central, not snRNA-vs-Visium (§5.6).

Artifact A3: harmonization report + final gene sets.

Step 4 — Place every cell (score + classifier)

Inputs: A2 + A3 (+ A1 training set for 4b).

4a — signature scoring (baseline + headline). Compute pericentral and periportal scores (Scanpy `score_genes` / our own background-subtracted mean) and define `zonation_coord = pericentral - periportal`, run with both the **transcriptome-wide** default and the **Paper 2 landmark** set, and confirm they agree.

4b — classifier (headline). Train on Paper 2's zone-labelled nuclei (calibrated multinomial logistic regression), evaluate by held-out accuracy / confusion matrix *on Paper 2 first*, then apply to Paper 1; record predicted zone, full zone-probability vector, and entropy per cell.

4c — agreement. Check the scoring coordinate and the classifier's expected-zone agree on cells (especially healthy).

Artifact A4: per-cell table [cell_id, donor, stage, zonation_coord, zone_probs, entropy] + score-coloured UMAP.

Step 5 — Validate the placement (positive control)

Input: A4 restricted to Paper 1's healthy donors.

In healthy liver zonation is intact, so the method must recover it: hepatocytes spread along a clear axis; GLUL/CYP2E1 rise with the coordinate and ASS1/HAL fall (quantify by Spearman correlation); the classifier is confident (low entropy). Recall (§5.7) that the healthy coordinate is a single broad hump — the proof is the marker slopes, not bimodality. If healthy cells do not separate, stop and fix scoring before touching disease data.

Artifact A5: validation figure + correlation/concordance values.

Step 5b — Is the ruler still a ruler? (the validity controls)

Input: A4 across all stages.

Per stage, run the four diagnostics of §5.6: (i) **internal coherence** — mean pairwise correlation among pericentral genes (and among periportal); (ii) **cross-program anti-correlation** — correlation between the two module scores across cells; (iii) **split-half reproducibility** — correlate two coordinates from random half-signatures; (iv) **program-off vs restriction-lost** — track each module's mean magnitude alongside the co-expression rate. Together these say *whether* and *how* the ruler degrades, so the Step-6 collapse is interpreted correctly rather than conveniently.

Artifact A5b: per-stage coherence / anti-correlation / split-half / decomposition plots.

Step 6 — Quantify the collapse across stages (headline, H1)

Inputs: A4 (all stages) + A5/A5b (validated).

6a — compute per donor, then per stage. The unit of inference is the **donor, not the cell** (§5.7). Compute each metric *per donor* — classifier **prediction-entropy** (headline), coordinate **spread, axis separability** (silhouette/AUC of the two ends), **marker anti-correlation** (5b) — then summarise across stages.

6b — trend test. Test the ordered Healthy → end-stage trend on the *per-donor* values (Jonckheere–Terpstra or Spearman on stage rank).

6c — error bars. Donor-level bootstrap for CIs — not cosmetic, but the thing that prevents pseudoreplication.

6d — negative control. Shuffle stage labels → the trend must vanish.

```
for d in donors:
    # UNIT = DONOR (~47), never cell
    spread[d] = std(coord[donor==d])
    anticorr[d] = spearman(pc[donor==d], pp[donor==d])
    entropy[d] = mean(cell_entropy[donor==d]) # if classifier run
    rank = stage_rank[donor_stage] # Healthy=0 ... End-stage=4
    rho,p = spearman(rank, spread) # ordered trend on ~47 values
    CI = bootstrap_over_DONORS(rho, B=2000) # error bars
```

```
p_perm = shuffle_stage_labels(rho, B=2000) # negative control
PASS if spread DOWN, anticorr -> 0, CI excludes 0, p_perm small
```

Artifact A6: the collapse curve + per-stage distributions + trend p-value (answers H1). Code: `src/steps/step6_collapse.py` · reference `pipeline.py:collapse` · plot `plot_a6_collapse`.



Fig. 8 — The headline deliverable (mock-up): a de-zonation metric rising monotonically across stages with donor-bootstrap CIs (one value per donor, tested across the ordered stages — see Step 6). Shape illustrative; the real curve is the result.

Step 7 — Which genes drive it (zone-resolved DE, FDR-controlled, H2)

Inputs: A4 + A2.

Bin hepatocytes into pseudo-zones (portal / mid / central terciles of the coordinate), then — to respect the donor as the unit of inference — **pseudobulk**: aggregate expression per **donor** × **zone** and test the stage effect on those profiles, BH-FDR across genes. A cell-level Wilcoxon/GLM looks powerful but is statistically invalid here (pseudoreplication, §5.7).

Avoiding circularity (important). The pseudo-zones were built from expression, so we must *not* then claim the signature genes themselves as "drivers." Three guards: (i) **exclude** the signature/classifier genes from driver claims (or label them validation-only); (ii) use a **held-out random gene split** — define the coordinate on one half of the zonation genes, test collapse on the other, and **repeat K≈20–50 times**, reporting the distribution across splits (a robust signal holds across most splits — meaningful only with the full ~1,640-gene set, not 20 landmarks); (iii) best, fit per gene expression ~ coordinate + stage + coordinate×stage — the **interaction term** ("this gene's zonal slope weakens with stage") is the honest "lost zonal pattern" test, not a re-test of the genes that built the axis.

```
zone = tercile(coord) # portal / mid / central
for z in [portal, central]:
    PB = pseudobulk(expr, by=(donor, zone)) # donor x gene mean log1p-CP10k
    for g in genes_detected_in(z): # >=10% of zone's cells
        fit: PB[g] ~ coord + stage + coord:stage # OLS on the donor aggregates
        p[g] = pvalue(coord:stage) # does the ZONAL SLOPE change with stage?
    q = BH(p)
report q<0.05 EXCLUDING signature genes # circularity guard (+ K repeated held-out splits)
```

Artifact A7: per-zone DE tables (gene, effect, p, q, is_signature) + volcano plots + the shortlist of earliest/strongest collapsing programs (answers H2). Code: `src/steps/step7_de.py` · reference `pipeline.py:de` · `plot` `plot_a7_volcano`.

Step 8 — Link de-zonation to plasticity (H3)

Inputs: A4 + plasticity markers (KRT7, KRT19, SOX9; + Paper 1 plasticity genes).

Score each hepatocyte for a cholangiocyte/plasticity signature (a *separate* axis from zonation, §2.4), then test the link **within donor** / **within stage** — *not* pooled across all cells, where late disease would trivially show more of both and fake a correlation (§5.7). Use per-donor correlations aggregated across donors, or a model `plasticity ~ de-zonation + stage + donor` (the de-zonation term is the real effect); a stage-stratified scatter makes the claim credible. Report effect size beside p.

Artifact A8: figure + statistic linking lost zonal identity to gained plasticity (answers H3).

6.4 Optional bonus (stretch) — does the collapse track inherited MASLD risk? (Paper 3)

The question, sharpened. Paper 3 (Nat Genet 2026) maps MASLD **inherited risk variants** to the genes they regulate (snATAC-seq + a massively-parallel reporter assay → a list of *risk-variant target genes*). We ask one precise thing: **are the genes whose zonal pattern collapses earliest (our Step-7 hits) over-represented among those inherited-risk target genes — more than chance?** A "yes" means the cell-state breakdown we measure and germline susceptibility **converge on the same molecular targets**; a "no" means de-zonation is mostly an *acquired*, downstream process. Either answer is publishable-grade context, and neither is needed for H1–H3.

The exact test (this is the whole method). It is a gene-set overlap test — light, no new heavy data:

```
hits          = Step7 genes with q<0.05 AND |interaction effect| large # collapse drivers
risk_genes    = Paper-3 risk-variant target genes (from its supplement)
background    = genes ACTUALLY TESTABLE in Step 7 (detected in the zone,
                survived pseudobulk) -- NOT all ~20k genes
# optionally expression-match the background to hits (same mean-expression bins)
k = |hits n risk_genes| ; draw = |hits| ; success = |background n risk_genes| ; N = |background|
p = hypergeom.sf(k-1, N, success, draw) # scipy.stats.hypergeom (or Fisher exact)
fold = (k/draw) / (success/N) # >1 = enrichment
report fold + 95% CI + BH-corrected p
```

Why the background matters (the easy way to fool yourself). Testing against all 20,000 genes guarantees fake enrichment, because both lists are biased toward highly-expressed liver genes. The honest background is the set of genes that *could* have been called in Step 7 (detected + tested), ideally expression-matched to the hit list. **Interpretation:** fold>1 with BH q<0.05 → genetics×cell-state convergence; fold≈1 → independent processes. **Caveats:** variant→gene mapping is imperfect; small hit lists give wide CIs; result is associational, not causal. **Code:** `src/steps/step9_bonus_enrichment.py` (~30 min, gene-list only). A second, even lighter stretch: Spearman-correlate the Step-6 collapse curve against a Paper 4 immune-infiltration trajectory across matched stages.

WHY IT'S A STRONG BONUS, NOT A DEPENDENCY

It is genuinely additive (germline genetics × somatic cell-state), cheap (one hypergeometric test you already know from Lecture 5's multiple-testing material), and strictly *optional* — the project stands fully on H1–H3 without it, so it can never sink the timeline.

6.5 Splitting the work between two people

Phase	Person A — reference & stats	Person B — disease data & modelling
Prep	Parse Paper 2; signatures (full + landmark) + training set (Step 1)	Download + QC Paper 1 hepatocytes (Step 2)
Day 1 AM	Harmonize, batch strategy (Step 3)	Scoring + classifier scaffold (Step 4)
Day 1 PM	Validation + ruler diagnostics (5, 5b)	Place all cells (Step 4 finished)
Day 2 AM	Collapse metrics + trend stats (Step 6)	Zone-binned DE + FDR (Step 7)
Day 2 PM	Plasticity link (Step 8) + figures	Write-up, reproducibility, stretch

6.6 Risk register

Risk	Mitigation
Platform batch dominates placement	Rank/z-scored features; train classifier on Paper 2's own snRNA-seq; validate on Paper 1 healthy cells (Step 5)
Ruler dissolves in disease	Step 5b diagnostics decide interpretation before any claim
.mat / .rds conversion eats time	Use the supplementary gene table for baseline signatures (no .mat); convert once to Python-native files then never revisit
Gene-ID mismatch	Explicit mapping + reported rate (A3)
Stage imbalance / pseudoreplication	Donor-level inference throughout (collapse per donor, pseudobulk donor×zone DE, within-donor plasticity); donor bootstrap + donor-label-shuffle control; report per-stage donor counts so imbalance is visible
Classifier sees out-of-distribution cells (disease nuclei unlike any healthy training cell)	Train on Paper 2 snRNA-seq (same platform); flag low-confidence/OOD cells; cross-check entropy against QC metrics and the signature score before calling it de-zonation
Entropy mis-read because probabilities are poorly calibrated	Prefer a calibrated multinomial logistic regression over a random forest; sanity-check calibration on the Paper 2 held-out split
Gene-set circularity (signature genes counted as "drivers")	Exclude signature/classifier genes from driver claims; K-repeated held-out gene splits for the coordinate vs the test; interaction-term test (Step 7)
	Use the corrected GEO version; auto-detect the donor column from several candidates; print donor counts per stage as a load-time check

Metadata / donor-label drift (e.g. the known GSM swap; donor-ID column ambiguity)

"Reproduction" perception

Foreground the cross-dataset coordinate, entropy collapse curve, and plasticity link

6.7 Statistical rigor checklist

- Significance always at **BH-FDR $q < 0.05$** , never raw p.
- Stage effect via an **ordered** trend test, not pairwise only.
- **Donor-level** resampling, never 69k cells treated as independent.
- DE is **pseudobulk** (donor $\times$ zone), with the H2 collapse tested via the **coordinate $\times$ stage interaction** on K-repeated held-out gene splits — not a re-test of the genes that built the axis.
- Coordinate computed both **transcriptome-wide** and on the **Paper 2 landmark** set; results shown to agree.
- A **positive control** (Step 5) and a **negative control** (shuffled labels).
- Classifier judged by held-out accuracy on Paper 2 before use on Paper 1.
- Effect sizes reported beside p/q.

6.8 What success looks like, and the artifacts

Success = a validated placement that recovers healthy zonation (A5), a monotonic collapse curve with honest CIs and a significant trend (A6), an FDR-controlled list of the programs that erode first (A7), and evidence that de-zonated hepatocytes are the plastic ones (A8) — with Step 5b telling us, honestly, what kind of breakdown each stage shows.

ID	Artifact	Answers
A1	Signatures (full + landmark) + zone-labelled training set + profile figure	reference
A2	QC'd normalized hepatocyte matrix + QC plots + counts	input
A3	Gene-harmonization report	batch control
A4	Per-cell coordinate + zone probs + entropy + UMAP	core decode
A5 / A5b	Healthy validation + ruler-validity diagnostics	method & ruler valid?
A6	Collapse curve + trend test	H1
A7	Zone-resolved DE tables (q) + volcanoes	H2
A8	De-zonation $\leftrightarrow$ plasticity link	H3
A9	Reproducible code repo + methods write-up	all

7 Timeline

When	Goal	Done when...
Prep	Data downloaded + converted; signatures (A1) + QC'd hepatocytes (A2) ready	Both datasets load in one notebook
Day 1	Harmonize (A3), place all cells (A4), validate (A5, A5b)	Method recovers zonation in healthy donors; score & classifier agree
Day 2	Collapse curve (A6), DE (A7), plasticity (A8), write-up (A9)	H1–H3 answered with controlled statistics

CRITICAL PATH

Step 5 must pass early. If the coordinate recovers healthy zonation by end of Day 1, the rest is mechanical — do not start the collapse analysis until the positive control passes.

MINIMUM VIABLE RESULT (THE FALLBACK IF TIME RUNS SHORT)

The project still fully succeeds on a reduced scope: the **signature coordinate** (Step 4a, no classifier needed) + the **healthy positive control** (Step 5) + the **donor-level collapse trend** for H1 (Step 6, spread & anti-correlation per donor) is a complete, defensible story on its own. Everything beyond — the classifier/entropy reader, the pseudobulk DE for H2, the plasticity link for H3, and the Paper 3 bonus — is layered on in that order, each independently droppable without invalidating what came before. Build H1 end-to-end first, then add upward.

8 Glossary

Term	Meaning
Hepatocyte / Cholangiocyte	Main metabolic liver cell / bile-duct cell (hepatocytes can drift toward this identity in disease)
Lobule	Hexagonal liver unit (0.5–1 mm)
Porto–central axis	Blood-flow gradient from portal inlet to central vein
Zonation / De-zonation	Position-dependent division of labour / its loss in disease (a population property, not one cell's coordinate)
Plasticity	Loss of cell-type identity (hepatocyte → cholangiocyte); a separate axis from zonation
Periportal / Pericentral	Relatively oxygen-rich (portal) / oxygen-poor (central) ends of the axis
MASLD / MASH	Fatty liver disease / steatohepatitis (fat + inflammation)
snRNA-seq	Single-nucleus RNA-seq; survives hard tissue, loses spatial coordinates, intron-heavy
Spatial transcriptomics	Expression measured with position kept (Visium / Visium HD / MERFISH / PhenoCycler)
UMI / Dropout	Counted unique molecule / a technical zero (gene on but missed)
QC	Quality control: discarding junk/low-quality cells before analysis
Signature score	Pooled expression of a gene set (robust to single-gene dropout)
Landmark / Transcriptome-wide program	Paper 2's anchor genes (baseline) / the full ~1,640 zoned genes (default); coordinate computed both ways
Zone classifier / entropy	Model predicting a cell's zone; flat (high-entropy) prediction = de-zoned
Zonation coordinate	Reconstructed 1D portal ↔ central position (pseudo-spatial, not x,y)
DE / FDR (Benjamini–Hochberg)	Testing which genes change / controlling false positives across many tests (q-value)
Pseudoreplication	Wrongly treating many cells from few donors as independent

9 References & data

1. **Paper 1.** Gribben C, Galanakis V, Calderwood A, et al. *Acquisition of epithelial plasticity in human chronic liver disease*. Nature (2024). doi:10.1038/s41586-024-07465-2. **Data:** GEO **GSE202379** (processed: `GSE202379_SeuratObject_AllCells.rds.gz`); code `github.com/Core-Bioinformatics/MASLD-NASH`.
2. **Paper 2.** Yakubovsky O, Bahar Halpern K, Shir S, et al. *A spatial atlas of the healthy human liver from live donors*. Nature (2026). doi:10.1038/s41586-026-10377-y. **Data:** Zenodo records **17735506** / **17735558** / **17735587** (the latter holds `zon_struct_all_full.mat` and `combined_scrnaseq_atlas_M5M6M7M8.mat`); code `github.com/OranYak/Human-liver`; browsable web app at `shalevapps.weizmann.ac.il/liver_app`. (Note: GSE248077/GSE223559/GSE240429 cited by this paper are *external comparison* datasets, not its own deposit.)
3. **Paper 3.** Zhu B, et al. *Integrative analyses ... risk alleles for metabolic liver disease*. Nature Genetics (2026). doi:10.1038/s41588-026-02617-8.
4. **Paper 4.** Martin OP, et al. *Single-cell atlas of human liver and blood immune cells across fatty liver disease stages*. Nature Immunology (2025). doi:10.1038/s41590-025-02255-y.
5. Benjamini Y, Hochberg Y. *Controlling the false discovery rate*. J. R. Statist. Soc. B (1995).
6. Halpern KB, Itzkovitz S, et al. *Single-cell spatial reconstruction reveals global division of labour in the mammalian liver*. Nature (2017) — the zonation-reconstruction concept.
7. Course materials, Computational Genomics 76553: Ex 1 (alignment), Ex 2 (library complexity), Lec 5 (multiple testing/FDR), Lec 15–16 (single-cell clustering).

Planning + study aid for the hackathon. Paper figures/tables are pointed to, not reproduced; verify exact supplementary file names and accession contents at download time.